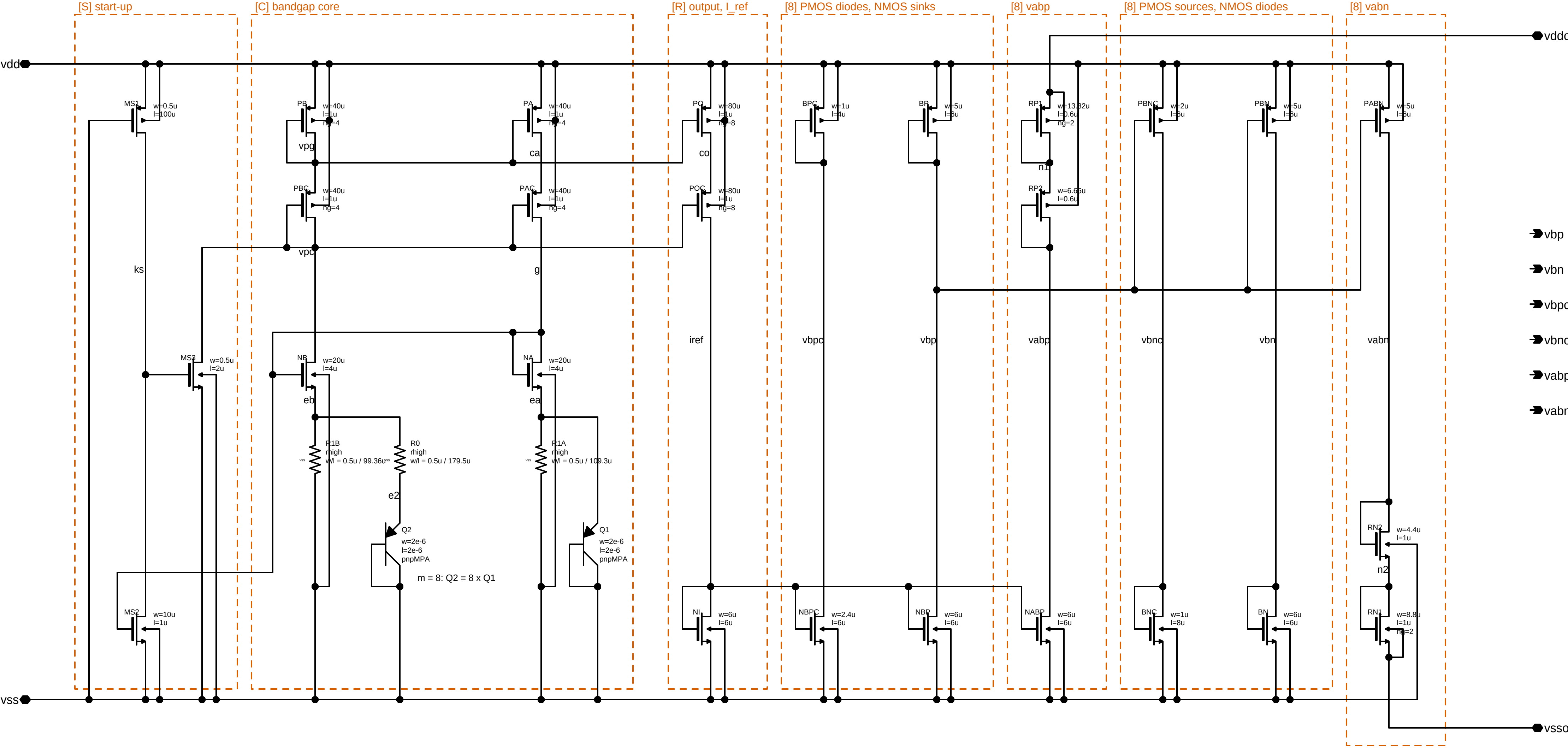


d2s_bias_bg: self-contained bias, current-mode bandgap with pnpMPA and rhigh + mirror tree of d2s_bias_in

branch current $V_{BE1}/R1 + U_T \ln 8 / R0 = 2.5 \text{ uA}$; $I_{ref} = 2 \times \text{branch} = 5 \text{ uA}$ into NI; Q1 / Q2: emitter $2u \times 2u$



bg_core: current-mode (Banba type) bandgap, no op-amp.

The PMOS mirror makes both branch currents equal and the NMOS pair NA / NB (gates on g) forces ea = eb, so each branch carries $V_{BE1} / R1 + U_T \ln 8 / R0$ ($Q2 = 8 \times Q1$). rhigh (-0.22 %/K) nearly cancels V_{BE} 's -1.77 mV/K already, so the PTAT share through R0 is small. PO / POC deliver twice the branch current, 5 uA, into NI. R1A is 10 % longer than R1B: with both PNPs off the loop gain is then > 1, so the core cannot rest there. PB / PBC are stacked diodes; the PMOS run in moderate inversion for headroom at 3.0 V, ss, -40 C.

Mirror tree and diodes. The six diodes are those of d2s_bias_lp and set the bias voltages:

vbp: BP 5u/6u, 5 uA (unit tails). vbpc: BPC 1u/4u, 2 uA (mirror cascodes). vabp: RP2 on RP1, 5 uA (ABP, FPL). vbn: BN 6u/6u, 5 uA (fold sinks). vbnc: BNC 1u/8u, 2 uA (fold cascodes). vabn: RN2 on RN1, 5 uA (ABN, FNL).

PMOS gate lines come from PMOS diodes on vdd, NMOS gate lines from NMOS diodes on vss: bias crosses blocks as currents.

RP1 and RN1 are the class-AB replicas of OP and ON: they sit on the output-stage rails vddo / vsso (separate rails from the right). RP2 keeps its n-well on vdd, like ABP. RN1 needs a local substrate tap ring on vsso, like ON.